

Math Whiz Cracks Wordle -- Wins 99% of the Time!

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Love Wordle? A team of math wizards just found a way to beat it 99% of the time! Instead of guessing common letters like A or E, they use a clever idea called **Shannon entropy**. It's like playing detective you pick words that tell you the most about the hidden answer, even if they seem random. Imagine you're trying to guess a secret five-letter word. Each guess gives you colored clues: green for right letter in the right spot, yellow for right letter in the wrong spot, and gray for wrong letter. The trick is to choose a guess that splits the remaining possibilities into the smallest groups. That way, you rule out tons of wrong words fast. The researchers tested their method against a normal strategy that picks popular letters. Their entropy-based approach solved puzzles in fewer tries and almost never lost. The project started as a class assignment! Professor Congyu Wu challenged his students to apply information theory to a real problem. They turned it into a published paper. So next time you play Wordle, remember: the best guess isn't always the most obvious one it's the one that teaches you the most.

Word Treasure

Shannon entropy -- A math idea that measures how much new information something gives you -- named after a famous scientist.

strategy -- A careful plan or method you use to try to win a game or solve a problem.

possibilities -- All the different words or answers that could still be the right one.

information theory -- A branch of math that studies how to send, measure, and use information in the smartest way.

assignment -- A task or project given by a teacher as part of schoolwork.

approach -- A way of doing something -- your chosen method for tackling a problem.

Background you need

Wordle is a viral online word game where you have six tries to guess a secret five-letter word; colored clues tell you if your letters are right or in the right place.

Claude Shannon (a brilliant American mathematician) invented information theory in the 1940s. His big idea -- that you can measure how much surprise or new information a message contains -- is used in everything from phone calls to wi-fi.

Shannon entropy helps you make the smartest choice by picking the option that rules out the most wrong answers at once. The Wordle team used this same principle to zoom in on the right answer faster than a normal guessing approach.

Break it down

WHO: A team of math students led by Professor Congyu Wu

WHAT: Built a Wordle-solving method using Shannon entropy that wins 99% of the time by picking guesses that reveal the most information

WHERE: Started as a classroom assignment and became a published scientific paper

WHY: To prove that information theory can crack real-world puzzles -- and that math is a powerful tool, not just schoolwork

Why it matters

This story shows that math isn't just about boring equations -- it's a secret weapon for solving puzzles you actually care about, from word games to big real-world problems.

Test what you remember

1. What clever idea do the math wizards use to beat Wordle?

- ☐ (A) Always guessing A and E first
- ☐ (B) Shannon entropy
- ☐ (C) Looking up words in a dictionary
- ☐ (D) Always starting with the same word

2. What does a green clue mean in Wordle?

- ☐ (A) Wrong letter, wrong spot
- ☐ (B) Right letter, wrong spot
- ☐ (C) Right letter, right spot
- ☐ (D) The game is over

3. How often does the entropy-based method win Wordle?

- ☐ (A) 50% of the time
- ☐ (B) 75% of the time
- ☐ (C) 99% of the time
- ☐ (D) 100% of the time

4. Who challenged students to apply information theory to a real problem?

- ☐ (A) A famous Wordle player on YouTube
- ☐ (B) Professor Congyu Wu
- ☐ (C) A popular science TV star
- ☐ (D) The person who invented Wordle

5. What did the Wordle project turn into?

- ☐ (A) A new video game
- ☐ (B) A published paper
- ☐ (C) A hit TV show
- ☐ (D) A new smartphone app

6. According to the article, what is the best guess in Wordle?

- ☐ (A) The most obvious common word
- ☐ (B) The word with the most vowels
- ☐ (C) The word that teaches you the most
- ☐ (D) The shortest five-letter word

Answer key (try the quiz first!): 1: B 2: C 3: C 4: B 5: B 6: C

Questions to think about

1. **Math as a superpower:** The students turned a fun game into a serious science project, proving that classroom math can do amazing things. Information theory helped them think like detectives instead of just lucky guessers.

2. Does winning take the fun away?: If a formula can beat Wordle almost every time, some players might feel the magic is gone. Part of the joy is the struggle -- but for these math wizards, cracking the code was the real thrill.

3. Learning beats luck: The best strategy wasn't about picking common letters -- it was about choosing words that teach you the most. That's a lesson for any puzzle, and maybe for life: curiosity beats guessing.

MY THOUGHTS (at least 20 words):